

First Law of Thermodynamics

5. Different Type of Thermodynamical Process and Use of First Law of Thermodynamics

Isochoric Process: When volume remain constant during the process the process is said to be isochoric process.

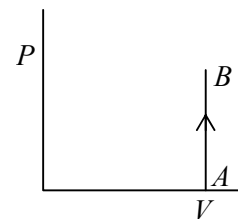
$$dW = PdV \quad dV = 0 \quad dW = 0$$

In an isochoric process work done during the process is zero.

From first law of thermodynamics.

$$dU = \delta Q - \delta W \quad \delta W = 0$$

$$dU = \delta Q = nC_v dT$$

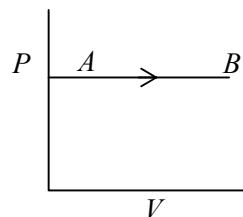


In isochoric process change in internal energy is equal to heat exchange.

Isobaric Process: When pressure of the system remain constant during the process.

Work done during the process is given by

$$\begin{aligned}\int_A^B P dV &= P(V_B - V_A) \\ &= nR(T_B - T_A) \\ &= nR\Delta T\end{aligned}$$



where ΔT is change in temperature during the process.

$$\delta Q = dU + \delta W = nC_V \Delta T + nR\Delta T = n(C_V + R)\Delta T$$

$$\delta Q = nC_P \Delta T \quad C_V + R = C_P \text{ (for the ideal gas)}$$

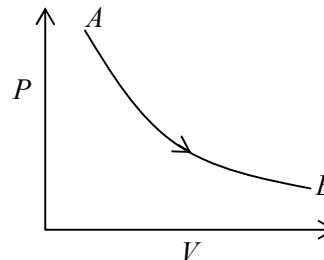
Heat exchange during the process for n mole of gas is equal to $nC_P \Delta T$ where C_P is specific heat capacity at constant process.

Isothermal Process: when temperature remains constant during the process then system is said isothermal process.

In isothermal process change in internal energy is zero because

$$\Delta T = 0$$

$$\delta W = PdV = nRT \int_{V_A}^{V_B} \frac{dV}{V} \quad \Rightarrow \quad \delta W = nRT \ln \frac{V_B}{V_A}$$



during the isothermal process heat exchange

$$\delta Q = \delta W = nRT \ln \frac{V_B}{V_A}$$

Adiabatic Process: When there is not any heat exchange during the process then process is said to be adiabatic process.

Adiabatic process is defined by

$$PV^\gamma = \text{constant}$$

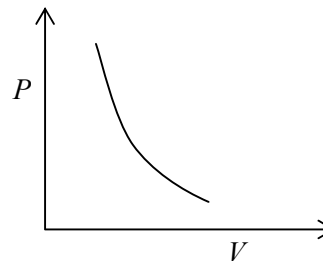
For Ideal gas $PV = RT$ for one mole of gas.

$$\text{So } TV^{\gamma-1} = \text{constant}$$

$$P^{1-\gamma} T^\gamma = \text{constant}$$

$$\text{where } \gamma = \frac{C_P}{C_V}$$

work done during adiabatic process $\delta W = PdV$



$$PV^\gamma = k \quad dW = \frac{k}{V^\gamma} dV = k \int_{V_A}^{V_B} V^{-\gamma} dV$$

$$k \frac{V^{-\gamma+1}}{-\gamma+1} \Big|_{V_A}^{V_B} = k \left(\frac{V_B^{1-\gamma} - V_A^{1-\gamma}}{1-\gamma} \right)$$

$$\frac{P_B V_B^\gamma V_B^{1-\gamma} - P_A V_A^\gamma V_A^{1-\gamma}}{1-\gamma} = \frac{P_B V_B - P_A V_A}{1-\gamma}$$

For n mole Ideal gas

$$W = \frac{nR(T_B - T_A)}{1-\gamma}$$

For Adiabatic process $\delta Q = dU + \delta W$

Hence $\delta Q = 0 \quad dU = -\delta W$

So in adiabatic process change in internal energy is equal to minus of work done i.e.

$$dU = \frac{nR(T_A - T_B)}{\gamma - 1}$$

Sign convention: Work done by the system is positive, work done on the system is negative, heat absorbed by system dQ is positive. Heat reject by the system is negative.

Example: n mole of certain ideal gas at temperature T_0 were cooled isochorically so that the gas pressure reduced n times. Then as a result at the isobaric process the gas expanded till its temperature get back to initial value. Find the total amount of heat absorbed by the gas in the process.

Solution: Let at state A, the pressure, volume, temperature is P_0, V_0, T_0, n is number of mole.

According to question:

$$\begin{array}{ccccc} \frac{P_0 V_0}{T_0} & \xrightarrow{\text{isochoric}} & \frac{P_0}{n} \frac{V_0}{T_B} & \xrightarrow{\text{isochoric}} & \frac{\frac{P_0}{n} V_c}{T_0} \\ \text{(A)} & & \text{(B)} & & \text{(C)} \end{array}$$

$$\frac{P_0 V_0}{T_0} = \frac{P_0 V_0}{n T_B} \quad T_B = \frac{T_0}{n}$$

$$\frac{P_0}{n} \frac{V_0}{T_B} = \frac{P_0}{n} \frac{V_c}{T_0} = V_c = n V_0$$

In process A to B

$$dQ = nC_v \Delta T = nC_v \left(\frac{T_0}{n} - T_0 \right)$$

In process B to C

$$dQ = nC_v \Delta T + PdV = nC_v \left(T_0 - \frac{T_0}{n} \right) + nR \left(T_0 - \frac{T_0}{n} \right)$$

Total heat absorbed by gas, $dQ(A \rightarrow B) + dQ(B \rightarrow C)$

$$= nC_v \left(\frac{T_0}{n} - T_0 \right) + nC_v \left(T_0 - \frac{T_0}{n} \right) + nR \left(T_0 - \frac{T_0}{n} \right) = nRT_0 \left(1 - \frac{1}{n} \right)$$

Example: A sample of an ideal gas is taken through the cyclic process $abca$ as shown in figure. It absorbed 50 J of heat during part ab , no heat during bc and reject 70 J of heat during ca , 40 J of work is done on the gas during part bc .

(a) Find the internal energy of the gas at b and c if it is 1500 J at a .

(b) Calculate the work done by the gas during the part ca .

Solution: (a) during $a \rightarrow b$ $V = \text{constant}$

$$\Delta Q = dU = 50J$$

$$U_a = 1500J \text{ so at } U_b = 1550J$$

Work $b \rightarrow c = -40 J$

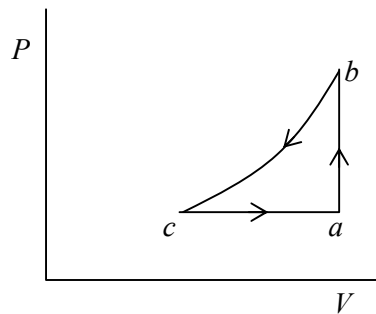
$$U_b = 1550J \quad U_c = 1590 \text{ (no heat at } bc)$$

(b) during path $c \rightarrow a$

$$\Delta U = 1500 - 1590 = -90J$$

$$\Delta Q = -70J$$

$$\Delta W = \Delta Q - \Delta U = -70 + 90 = 20J$$

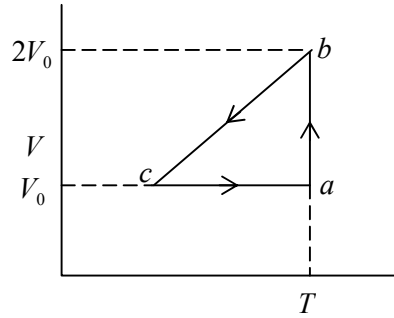


Example: A sample of an Ideal gas has pressure P_0 , volume V_0 and temperature T_0 . It is isothermally expand to twice its original volume. It is then compressed at constant pressure to have the original volume V_0 . Finally, the gas is heated at constant volume to get the original temperature.

(a) Show the process in VT diagram.

(b) Calculate the heat absorbed in the process.

Solution: (a)



(b) Hence the process is cyclic then change in internal energy in cycle is zero. So heat supplied in the cycle is equal to work done.

Work done during A $\rightarrow$ B is isothermal, so

$$W_{AB} = nRT_0 \ln \frac{2V_0}{V_0} = nRT_0 \ln 2$$

b $\rightarrow$ c isobaric, so work done is PdV .

$$P_a V_a = P_b V_b$$

$$P_0 V_{0a} = P_b 2V_0$$

$$P_b = \frac{P_0}{2}$$

$$W_{b \rightarrow c} = \frac{P_0}{2} (V_0 - 2V_0) = -\frac{P_0 V_0}{2}$$

Wc $\rightarrow$ a is isochoric Wc $\rightarrow$ a = 0

Total work done is given by

$$nRT_0 \ln 2 + \left(-\frac{P_0 V_0}{2} \right) = nRT_0 \left(\ln 2 - \frac{1}{2} \right).$$